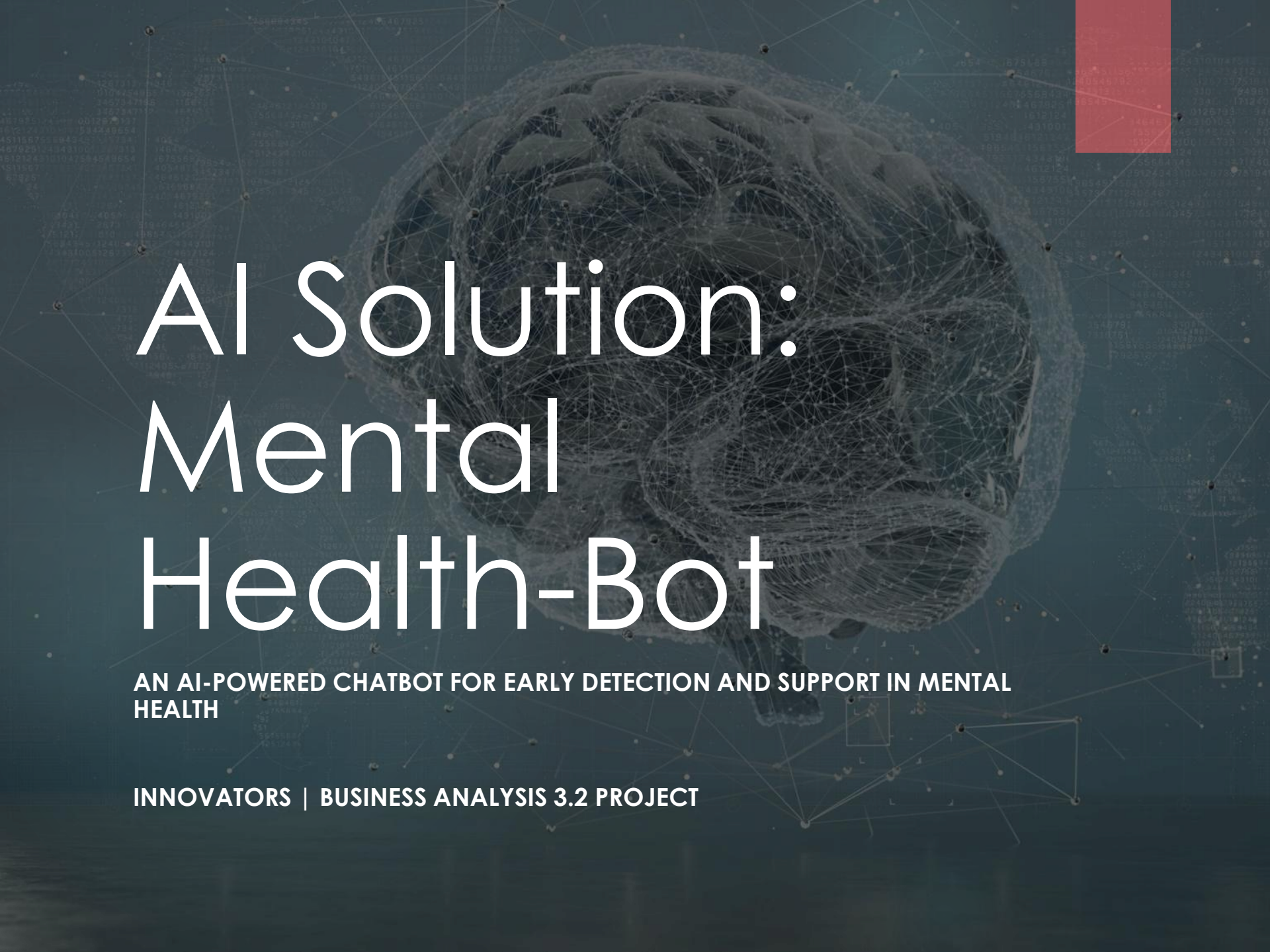




AI Solution: Mental Health-Bot

AN AI-POWERED CHATBOT FOR EARLY DETECTION AND SUPPORT IN MENTAL HEALTH

INNOVATORS | BUSINESS ANALYSIS 3.2 PROJECT

Introduction

- ▶ The Fourth Industrial Revolution (4IR) has created new opportunities for technology to solve societal problems. Despite global progress, developing nations still face limited access to mental health resources. This project introduces an AI-driven Mental Health Chatbot that leverages NLP and Machine Learning to detect symptoms and guide users toward suitable care.

Global Perspective



**MENTAL HEALTH
DISORDERS AFFECT OVER 1
BILLION PEOPLE
WORLDWIDE.**



**75% OF PEOPLE IN LOW-
INCOME COUNTRIES
RECEIVE NO TREATMENT.**



**THE COVID-19 PANDEMIC
AND ABUSE WORSENE
MENTAL HEALTH
CONDITIONS GLOBALLY.**



**AI OFFERS SCALABLE,
AFFORDABLE, AND
CONFIDENTIAL SUPPORT
SOLUTIONS.**

Common Mental Health Disorders



**DEPRESSION –
PERSISTENT SADNESS,
LOSS OF INTEREST.**



**ANXIETY – EXCESSIVE
WORRY, RESTLESSNESS.**



**STRESS – CHRONIC
TENSION LEADING TO
BURNOUT.**



**PANIC DISORDER –
SUDDEN EPISODES OF
FEAR OR DISCOMFORT.**

Causes and Risk Factors



Socioeconomic stress and unemployment.



Genetic predisposition and family history.



Sleep deprivation and poor diet.



Social isolation and stigma.

Impact on Daily Life

- ▶ Untreated mental health issues lead to reduced productivity, relationship strain, and, in severe cases, suicidal behavior. Communities and municipalities face healthcare overloads and economic losses due to undiagnosed conditions.

Mental Health Stigma

Cultural taboos
prevent open
discussion.

Fear of judgment
delays treatment.

AI chatbots offer
anonymity,
encouraging
individuals to seek
help.

Treatment and Support

1

Professional therapy and counseling remain critical.

2

AI chatbots can provide early screening and recommendations.

3

High-risk users are referred to professionals or emergency contacts.

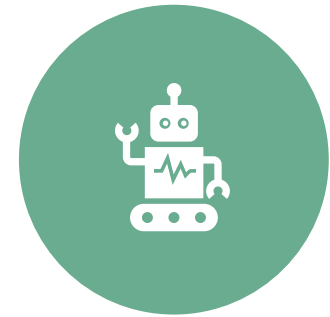
Promoting Mental Well-being



ENCOURAGE DAILY SELF-CARE AND MINDFULNESS.



INCREASE ACCESS TO AFFORDABLE DIGITAL HEALTH TOOLS.



USE AI TO PERSONALIZE USER FEEDBACK AND LEARNING.

Data Insights



DATASET:
FINAL_DEPRESSION_DATAS
ET_1.CSV WITH LIFESTYLE
AND MENTAL HEALTH
FEATURES.



TARGET: DEPRESSION
(YES/NO).



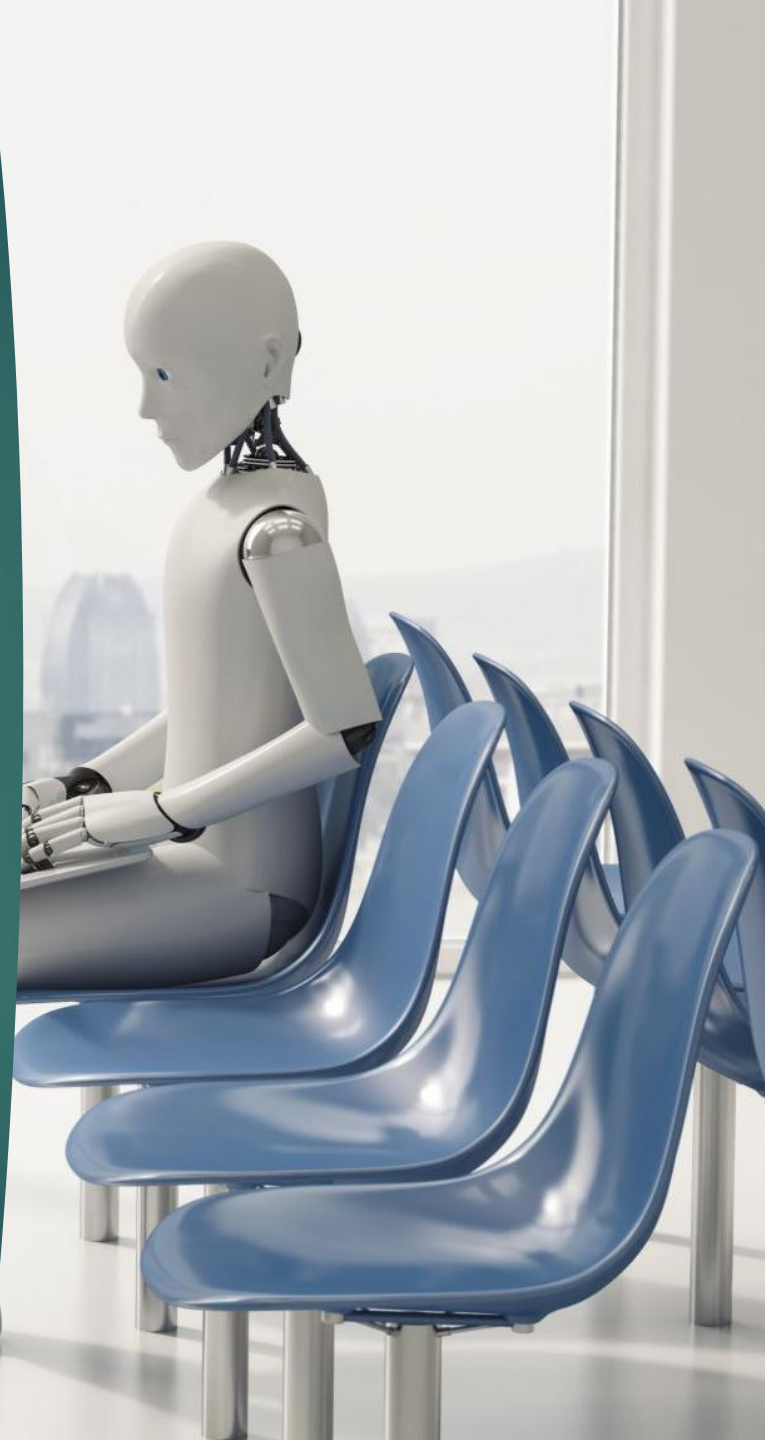
MODEL: RANDOM FOREST
WITH 96% ACCURACY.



NLP TECHNIQUES:
SENTIMENT ANALYSIS,
EMOTION DETECTION, AND
TEXT CLASSIFICATION.

Conclusion

- ▶ The Mental Health-Bot demonstrates how AI can bridge healthcare gaps by offering early mental health detection, awareness, and support. It empowers individuals, reduces stigma, and supports municipalities in managing community well-being through 4IR technologies.





Thank
you